

Report on Faculty Development Program (FDP)

Title: *Analysis of Recent Trends in Electric Vehicles, Battery Technology, and Integrated Control Systems*

Organized by: Institute of Engineering & Management, Salt Lake, Kolkata

Dates: 13th May – 17th May 2025

Technically Co-Sponsored by: IEEE CSS-IMS Joint Chapter, IEEE Kolkata Section

The Faculty Development Program (FDP) was conducted to impart in-depth knowledge of emerging technologies in electric vehicles (EVs), battery systems, and integrated control strategies. It aimed to empower faculty, researchers, and industry professionals with the latest developments, hands-on perspectives, and interdisciplinary insights. The event witnessed participation from over 120 individuals including faculty members, IEEE members, and industry delegates.

Day-Wise Summary of Sessions

Day 1: 13th May 2025

1. **Dr. Raghu Selvaraj** (Senior Scientist, CMERI, Durgapur)
Topic: *Recent Development of Electric Vehicles and Battery Technology*
Time: 10:00 AM – 11:30 AM
A technical overview of the state-of-the-art innovations in EV and battery domains.
2. **Dr. Asok Adak** (Associate Dean, IEST Shibpur)
Topic: *An Overview and Analysis of the New Educational Policy*
Time: 2:00 PM – 3:30 PM
Discussed the impact of the New Education Policy on engineering curricula.

Day 2: 14th May 2025

3. **Dr. Chandan Kumar Ghosh** (Jadavpur University)
Topic: *Pioneering Fabrication Processes for Next-Generation Energy Solutions*
Time: 10:00 AM – 11:30 AM
Focused on innovative materials and fabrication methods in energy technology.
4. **Dr. Ranjan Kumar Behera** (Associate Professor, IIT Patna)
Topic: *High-Performance Electrical Drives for Traction and EV Applications*
Time: 2:00 PM – 3:30 PM
Explained drive system designs for efficiency in EV powertrains.

Day 3: 15th May 2025

5. **Dr. Koushik Mukherjee** (IEST Shibpur)
Topic: *Charging and Discharging of Electric Vehicles*
Time: 10:00 AM – 11:30 AM
Explored EV battery operation, charging algorithms, and SOC estimation.
 6. **Dr. Rajanand Patnaik Narasipuram** (Chief Architect, Cyient)
Topic: *Architectural Frameworks for Real-Time Operation in Electric Vehicles*
Time: 2:00 PM – 3:30 PM
Presented real-time system design principles for EV performance monitoring.
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Day 4: 16th May 2025

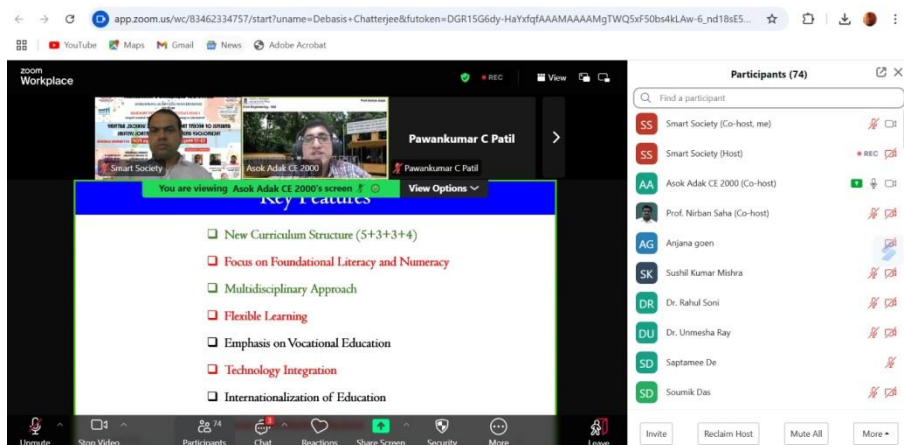
7. **Mrs. Malini Chaudhury** (Art of Living Faculty)
Topic: *Mental Health and Stress Management*
Time: 10:00 AM – 11:30 AM
Shared mindfulness practices and techniques to promote mental well-being.
8. **Dr. Amrita Sengupta** (Battery Modeling Engineer, Agratas)
Topic: *Next-Generation Battery Modeling: A Gateway to Efficient Energy Solutions*
Time: 2:00 PM – 3:30 PM
Demonstrated how modeling enhances battery design and energy optimization.

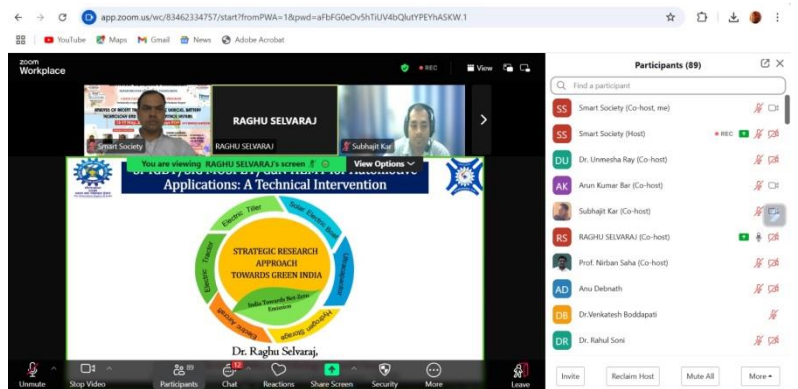
Day 5: 17th May 2025

9. **Dr. Chiranjit Sain** (Assistant Professor, GKCIET, Malda)
Topic: *Smart Electric Mobility Solutions for Sustainable Transportation*
Time: 10:00 AM – 11:30 AM
Focused on smart grid integration and intelligent transportation systems.

This FDP effectively bridged the gap between academia, research, and industry through in-depth technical sessions and interdisciplinary insights. With participation from nine eminent experts across institutions and industries, the event provided attendees with a future-ready understanding of electric vehicles and battery technologies. The program also integrated wellness and policy-oriented discussions, ensuring a holistic professional development experience. Certificates were distributed based on assessment test marks and adequate attendance percentage.

Some Pictures are added below:







INSTITUTE OF ENGINEERING & MANAGEMENT

DEPARTMENT OF ELECTRICAL ENGINEERING

PRESENTS

5 DAYS FACULTY DEVELOPMENT PROGRAM

Technically co-sponsored by IEEE Joint CSS-IMS Kolkata Chapter

ANALYSIS OF RECENT TRENDS IN ELECTRIC VEHICLES, BATTERY TECHNOLOGY AND INTEGRATED CONTROL SYSTEMS

HYBRID MODE

13-17 May, 2025

5 Days FDP

EMINENT SPEAKERS



Dr. Raghu Selvaraj
CMERI, Durgapur



Dr. Kaushik Mukherjee
IEST Shibpur



Dr. Ranjan Kumar
Behera, IIT Patna



Dr. Chiranjit Sain
GKCIET, West Bengal



Dr. Amrita Sengupta
Engineer, Agratas



Dr. Chandan Kr Ghosh
Jadavpur University



Dr. Rajanand Patnaik
Narasipuram, Cyient Ltd.



Mrs. Malini Chaudhury
Consultant, SSDYPCE



Dr. Asok Adak
IEST Shibpur

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**** 80% Attendance and 60% marks in Exam is
mandatory for certification**



Venue: D1, Management House, IEM Kolkata (Salt Lake Campus), Kol-91

